

Intro to Analysis

MATH 147

(written in the style of Zorich)

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1 The Real Numbers

Differential calculus studies slopes and tangent lines; integral calculus studies areas and accumulation. Both are built on the real numbers, and the purpose of this chapter is to say exactly what the real numbers are and which of their properties analysis actually uses.

1.1 Numbers and the Language of Logic

a. The number systems

We start from the familiar systems

$$\begin{aligned}\mathbb{N} &= \{1, 2, 3, \dots\}, \\ \mathbb{Z} &= \{\dots, -1, 0, 1, \dots\}, \\ \mathbb{Q} &= \left\{ \frac{m}{n} : m, n \in \mathbb{Z}, n \neq 0 \right\}.\end{aligned}$$

Each one repairs a defect of the previous one. $\mathbb{N}$ is closed under addition; $\mathbb{Z}$ is also closed under subtraction and multiplication, but not under division ($1/2 \notin \mathbb{Z}$); $\mathbb{Q}$ admits all quotients with nonzero denominator. Even $\mathbb{Q}$, however, is too small: it has no number whose square is 2.

Theorem 1.1.1 (Irrationality of $\sqrt{2}$)
 $\sqrt{2} \notin \mathbb{Q}$.

The idea is to write $\sqrt{2}$ in lowest terms; the equation $m^2 = 2n^2$ then forces both m and n to be even.

Proof

Suppose $\sqrt{2} = m/n$ with $m, n \in \mathbb{Z}$, $n > 0$, $\gcd(m, n) = 1$. Squaring gives $m^2 = 2n^2$, so m^2 is even. Since an odd integer has an odd square, m is even, say $m = 2\ell$. Then $4\ell^2 = 2n^2$, i.e. $n^2 = 2\ell^2$, so n is even by the same argument. Thus 2 divides both m and n , contradicting $\gcd(m, n) = 1$. ■

The elements of $\mathbb{R} \setminus \mathbb{Q} = \{x \in \mathbb{R} : x \notin \mathbb{Q}\}$ are called *irrational numbers*.

b. Decimal representation

Informally, every nonnegative real number has a decimal expansion

$$x = a + \sum_{i=1}^{\infty} \frac{x_i}{10^i}, \quad a \in \mathbb{N} \cup \{0\}, \quad x_i \in \{0, 1, \dots, 9\},$$

and negative numbers are negatives of such expansions. The expansion need not be unique.

Example 1.1.2

$1.999\ldots = 2$. Indeed, if $x = 1.999\ldots$, then $10x = 19.999\ldots$, so $9x = 10x - x = 18$ and $x = 2$.

This description of $\mathbb{R}$ is convenient but not a foundation: it does not tell us what an infinite sum is. In §1.3 we replace it by a list of axioms.

c. Statements and quantifiers

A *statement* is a sentence that is either true or false. We use the *quantifiers* $\forall$ (“for all”) and $\exists$ (“there exists”). Each quantifier is written in parentheses, and the statement it governs in square brackets. For example,

$$\begin{aligned} (\forall x \in \mathbb{R}) [x^2 - x \geq 0] & \text{ is false (take } x = \tfrac{1}{2}), \\ (\exists x \in \mathbb{R}) [x^2 - x \geq 0] & \text{ is true (take } x = 2). \end{aligned}$$

Negation interchanges the quantifiers:

$$\neg[(\forall x) P(x)] \iff (\exists x) [\neg P(x)], \quad \neg[(\exists x) P(x)] \iff (\forall x) [\neg P(x)].$$

The *order* of quantifiers matters, as the next two propositions show.

Proposition 1.1.3

$$(\forall x \in \mathbb{R})(\exists y \in \mathbb{R}) [x^3 - y^3 = 1].$$

Proof

Given x , take $y = \sqrt[3]{x^3 - 1}$; then $x^3 - y^3 = 1$. ■

Proposition 1.1.4

The statement $(\exists y \in \mathbb{R})(\forall x \in \mathbb{R}) [x^3 - y^3 = 1]$ is false.

Proof

We prove its negation $(\forall y \in \mathbb{R})(\exists x \in \mathbb{R}) [x^3 - y^3 \neq 1]$. Given y , take $x = y$; then $x^3 - y^3 = 0 \neq 1$. ■

Remark

In Proposition 1.1.3 the number y may depend on x , and it does: $y = \sqrt[3]{x^3 - 1}$. The second statement demands a single y that works for every x at once, which is much stronger. Interchanging $\forall$ and $\exists$ can turn a true statement into a false one.

1.2 The Natural Numbers: Induction and Well-Ordering

Two properties of $\mathbb{N}$ are used throughout the course. Induction lets us prove a statement for every n by passing from n to $n + 1$; well-ordering lets us pick the least element of a set of natural numbers. We take the first as a principle and derive the second from it.

a. Mathematical induction**Axiom** (Principle of Mathematical Induction)Let $P(n)$ be a statement depending on $n \in \mathbb{N}$. If

- (1) $P(1)$ is true, and
- (2) for every $k \in \mathbb{N}$, $P(k)$ implies $P(k + 1)$,

then $P(n)$ is true for every $n \in \mathbb{N}$.

Thus an induction proof has two parts: the *base case* $P(1)$, and the *inductive step*, in which we fix an arbitrary k , assume $P(k)$ (the *inductive hypothesis*), and deduce $P(k + 1)$.

Example 1.2.1

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \text{ for every } n \in \mathbb{N}.$$

ProofFor $n = 1$ both sides equal 1. If the formula holds for k , then

$$1 + \cdots + k + (k + 1) = \frac{k(k+1)}{2} + (k + 1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2},$$

which is the formula for $k + 1$. ■

The same principle, stated for sets, reads as follows.

Proposition 1.2.2If $S \subseteq \mathbb{N}$, $1 \in S$, and $k \in S \Rightarrow k + 1 \in S$, then $S = \mathbb{N}$.**Proof**Apply induction to the statement $P(n)$: “ $n \in S$ ”. ■**b. The well-ordering principle****Theorem 1.2.3** (Well-Ordering Principle)Every nonempty subset S of $\mathbb{N}$ has a least element.

Suppose S had no least element. We show by induction that then S contains none of $1, 2, \dots, n$, for every n ; hence $S = \emptyset$.

ProofAssume $S \neq \emptyset$ has no least element, and let

$$T = \{n \in \mathbb{N} : \{1, 2, \dots, n\} \cap S = \emptyset\}.$$

$1 \in T$: otherwise $1 \in S$, and 1, being the least natural number, would be the least element of S .

$k \in T \Rightarrow k + 1 \in T$: if $\{1, \dots, k\} \cap S = \emptyset$ but $\{1, \dots, k + 1\} \cap S \neq \emptyset$, then $k + 1 \in S$ while no smaller number lies in S , so $k + 1$ would be the least element of S .

By Proposition 1.2.2, $T = \mathbb{N}$. Hence every $n \in \mathbb{N}$ satisfies $n \notin S$, i.e. $S = \emptyset$, a contradiction. ■

Well-ordering gives the method of the *minimal counterexample*: if $P(n)$ failed for some n , the set $S = \{n \in \mathbb{N} : \neg P(n)\}$ would be nonempty and would have a least element k . Then $P(k - 1)$ holds by minimality, and one shows that $P(k - 1)$ forces $P(k)$ — a contradiction.

Example 1.2.4

Using well-ordering, prove that for every $n \in \mathbb{N}$

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}. \quad (1.2.1)$$

Proof

Let $P(n)$ be the statement (1.2.1), and suppose that the set of counterexamples

$$S = \{n \in \mathbb{N} : \neg P(n)\}$$

is nonempty. By Theorem 1.2.3, S has a least element k . Since $P(1)$ holds, $k > 1$, so $k - 1 \in \mathbb{N}$, and $P(k - 1)$ holds because $k - 1 < k$:

$$1^2 + \dots + (k - 1)^2 = \frac{(k - 1)k(2k - 1)}{6}.$$

Adding k^2 ,

$$\begin{aligned} 1^2 + \dots + k^2 &= \frac{k((k - 1)(2k - 1) + 6k)}{6} \\ &= \frac{k(2k^2 + 3k + 1)}{6} = \frac{k(k + 1)(2k + 1)}{6}, \end{aligned}$$

so $P(k)$ holds, contradicting $k \in S$. Hence $S = \emptyset$. ■

Remark

The argument needs the *least element* k of S , not just a greatest lower bound: we must know that $k \in S$, i.e. that $P(k)$ fails. For a nonempty subset of $\mathbb{N}$ this is exactly what well-ordering provides; for a subset of $\mathbb{R}$ the infimum need not belong to the set (§1.4).

1.3 The Axioms for the Real Numbers

The real numbers form a *complete ordered field*. The axioms come in three groups: the *field axioms* govern $+$ and $\cdot$ (they let us compute), the *order axioms* govern $<$ (they let us compare), and the *completeness axiom* is what makes limits possible. The first two groups hold in $\mathbb{Q}$ as well; only completeness separates $\mathbb{R}$ from $\mathbb{Q}$.

a. The field axioms**Definition 1.3.1**

A *field* is a set F with two operations $+, \cdot : F \times F \rightarrow F$ such that for all $a, b, c \in F$:

- F1.** $a + (b + c) = (a + b) + c$;
- F2.** $a + b = b + a$;
- F3.** there is $0 \in F$ with $a + 0 = a$;
- F4.** for each a there is $-a \in F$ with $a + (-a) = 0$;
- F5.** $a(bc) = (ab)c$;
- F6.** $ab = ba$;
- F7.** there is $1 \in F$, $1 \neq 0$, with $a \cdot 1 = a$;
- F8.** for each $a \neq 0$ there is $a^{-1} \in F$ with $aa^{-1} = 1$;
- F9.** $a(b + c) = ab + ac$.

$\mathbb{Q}$ is a field, so these axioms say nothing about size or order. What they do is justify every routine algebraic manipulation. Even “obvious” facts must be derived from them:

Theorem 1.3.2

Let $a, b, u, z \in \mathbb{R}$.

- (a) If $z + a = a$, then $z = 0$.
- (b) If $b \neq 0$ and $ub = b$, then $u = 1$.
- (c) $a \cdot 0 = 0$.

Proof

- (a) $z = z + 0 = z + (a - a) = (z + a) - a = a - a = 0$.
- (b) $u = u \cdot 1 = u(bb^{-1}) = (ub)b^{-1} = bb^{-1} = 1$.
- (c) $a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0$; adding $-(a \cdot 0)$ to both sides gives $0 = a \cdot 0$. ■

b. The order axioms**Definition 1.3.3**

There is a nonempty set $P \subseteq \mathbb{R}$, the *positive real numbers*, such that

- O1.** (*trichotomy*) for every $a \in \mathbb{R}$ exactly one of $a = 0$, $a \in P$, $-a \in P$ holds;
- O2.** $a, b \in P \Rightarrow a + b \in P$;
- O3.** $a, b \in P \Rightarrow ab \in P$.

We write $a > b$ (or $b < a$) if $a - b \in P$, and $a \geq b$ (or $b \leq a$) if $a - b \in P \cup \{0\}$.

Every inequality is thus a statement about membership in P , and every manipulation of inequalities reduces to O1–O3. Trichotomy is what makes case analysis possible, as in the following example.

Example 1.3.4

If $a \neq 0$, then $a^2 > 0$.

Proof

By trichotomy, $a \in P$ or $-a \in P$. If $a \in P$, then $a^2 = a \cdot a \in P$ by O3. If $-a \in P$, then $(-a)(-a) \in P$ by O3, and $(-a)(-a) = a^2$: multiplying $a + (-a) = 0$ by $-a$ gives $(-a)a + (-a)(-a) = 0$, so $(-a)(-a) = -((-a)a) = -(-a^2) = a^2$. ■

The positive rationals also satisfy O1–O3, so $\mathbb{Q}$ is an ordered field too. What it lacks is completeness (§1.5).

c. Absolute value**Definition 1.3.5**

The *absolute value* of $a \in \mathbb{R}$ is $|a| = a$ if $a \geq 0$ and $|a| = -a$ if $a < 0$.

Geometrically, $|a - b|$ is the distance between a and b on the line.

Theorem 1.3.6

For all $a, b \in \mathbb{R}$ and $c > 0$:

- (a) $|ab| = |a| |b|$;
- (b) $|a|^2 = a^2$;
- (c) $|a| \leq c \iff -c \leq a \leq c$;
- (d) $-|a| \leq a \leq |a|$.

Proof

Each part follows by splitting into the cases $a \geq 0$ and $a < 0$. Parts (b) and (d) are immediate from the definition.

(a) If $a, b \geq 0$, then $ab \geq 0$ and $|ab| = ab = |a| |b|$. If $a \geq 0 > b$, then $ab \leq 0$ and $|ab| = -ab = a(-b) = |a| |b|$; the case $b \geq 0 > a$ is symmetric. If $a, b < 0$, then $ab > 0$ and $|ab| = ab = (-a)(-b) = |a| |b|$.

(c) Let $|a| \leq c$. If $a \geq 0$, then $a = |a| \leq c$ and $-c < 0 \leq a$. If $a < 0$, then $-a = |a| \leq c$, i.e. $-c \leq a$, and $a < 0 < c$. Conversely, let $-c \leq a \leq c$. If $a \geq 0$, then $|a| = a \leq c$; if $a < 0$, then $-c \leq a$ gives $|a| = -a \leq c$. ■

Part (c) converts one inequality for $|x|$ into two ordinary inequalities for x and back; it will be used constantly. Its first application:

Proposition 1.3.7 (Triangle inequality)

$|a + b| \leq |a| + |b|$ for all $a, b \in \mathbb{R}$.

Proof

By Theorem 1.3.6(d), $-|a| \leq a \leq |a|$ and $-|b| \leq b \leq |b|$. Adding,

$$-(|a| + |b|) \leq a + b \leq |a| + |b|.$$

If $|a| + |b| = 0$, then $a + b = 0$ and there is nothing to prove. Otherwise Theorem 1.3.6(c), with $c = |a| + |b|$, gives $|a + b| \leq |a| + |b|$. ■

Corollary 1.3.8

For all $a, b \in \mathbb{R}$:

- (1) $||a| - |b|| \leq |a - b|$;
- (2) $|a - b| \leq |a| + |b|$.

Proof

(1) Apply the triangle inequality to $a = (a - b) + b$ and to $b = (b - a) + a$:

$$|a| - |b| \leq |a - b|, \quad |b| - |a| \leq |b - a| = |a - b|.$$

Hence $-|a - b| \leq |a| - |b| \leq |a - b|$, which is (1) by Theorem 1.3.6(c).

$$(2) |a - b| = |a + (-b)| \leq |a| + |-b| = |a| + |b|.$$

■

d. The completeness axiom

We state the third axiom now. The notions it uses are made precise in §1.4, and its consequences are the subject of §1.5.

Axiom (Completeness)

Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound (supremum) in $\mathbb{R}$.

$\mathbb{Q}$ does not have this property: the set $\{q \in \mathbb{Q} : q^2 < 2\}$ is bounded above in $\mathbb{Q}$ but has no least upper bound in $\mathbb{Q}$ (see Proposition 1.5.9).

1.4 Bounded Sets; Supremum and Infimum

Bounds make sense in any ordered field. What is special about $\mathbb{R}$ is that a set bounded above always has a *least* upper bound in $\mathbb{R}$; this section introduces the language needed to say so.

Definition 1.4.1

A nonempty set $S \subseteq \mathbb{R}$ is

- (1) *bounded above* if there is $u \in \mathbb{R}$ (an *upper bound*) with $x \leq u$ for all $x \in S$;
- (2) *bounded below* if there is $w \in \mathbb{R}$ (a *lower bound*) with $w \leq x$ for all $x \in S$;
- (3) *bounded* if it is bounded above and below, and *unbounded* otherwise.

Definition 1.4.2

An upper bound u_0 of S is called the *supremum* (least upper bound) of S , written $u_0 = \sup S$, if $u_0 \leq u$ for every upper bound u of S .

A lower bound w_0 of S is called the *infimum* (greatest lower bound) of S , written $w_0 = \inf S$, if $w_0 \geq w$ for every lower bound w of S .

The notation $\sup S$ is justified because a set has at most one supremum: if u_1, u_2 are both suprema, then $u_1 \leq u_2$ (as u_2 is an upper bound and u_1 is least) and likewise $u_2 \leq u_1$. The same holds for the infimum.

Checking that u is the *least* upper bound requires comparing u with every other upper bound. The following lemma replaces this by a condition on S itself: no matter how small a margin ϵ we allow, S has a point within ϵ of u .

Lemma 1.4.3 (ϵ -characterization of the supremum)

An upper bound u of a nonempty set $S \subseteq \mathbb{R}$ is the supremum of S if and only if

$$(\forall \epsilon > 0)(\exists x_\epsilon \in S) [u - \epsilon < x_\epsilon].$$

Proof

($\Rightarrow$) Let $u = \sup S$ and $\epsilon > 0$. Since $u - \epsilon < u$ and u is the least upper bound, $u - \epsilon$ is not an upper bound, so some $x_\epsilon \in S$ satisfies $u - \epsilon < x_\epsilon$.

($\Leftarrow$) If u were not least, there would be an upper bound $w < u$. For $\epsilon = u - w > 0$ we get $x_\epsilon \in S$ with $w = u - \epsilon < x_\epsilon$, so w is not an upper bound after all. ■

Thus, to prove $u = \sup S$ one checks two things: (1) u is an upper bound of S ; (2) for every $\epsilon > 0$ some $x \in S$ satisfies $u - \epsilon < x \leq u$. For $\inf S$ reverse the inequalities.

Example 1.4.4

Let $S = \{x \in \mathbb{R} : 0 \leq x < 1\}$. Then $\sup S = 1$ and $\inf S = 0$.

Proof

$\sup S = 1$. Clearly 1 is an upper bound. By completeness $u = \sup S$ exists, and $u \leq 1$. Suppose $u < 1$ and put $\epsilon = \frac{1-u}{2} > 0$. Then

$$u < u + \epsilon = \frac{u+1}{2} < 1,$$

and $u + \epsilon \geq 0$ because $u \geq 0$ (as $0 \in S$). So $u + \epsilon \in S$ exceeds the upper bound u , a contradiction. Hence $u = 1$.

$\inf S = 0$. Clearly 0 is a lower bound. Every lower bound w satisfies $w \leq 0$ because $0 \in S$. Hence 0 is the greatest lower bound. ■

Remark

In this example $\sup S = 1 \notin S$, so S has no largest element, whereas $\inf S = 0 \in S$ is also its least element. A supremum is a *maximum* exactly when it belongs to the set. Note

also that “bounded” is a property of the set S , while $\sup S$ and $\inf S$ are numbers attached to it.

1.5 Consequences of the Completeness Axiom

Every result of this section is proved by the same device: form the supremum of a well-chosen set (the completeness axiom guarantees it exists) and play the two defining properties of the supremum against each other.

a. Infima

Corollary 1.5.1

Every nonempty subset of $\mathbb{R}$ that is bounded below has an infimum in $\mathbb{R}$.

The idea is to reflect the set through 0: lower bounds of S become upper bounds of $-S$.

Proof

Let w be a lower bound of S and put $S' = \{-s : s \in S\}$. Since $w \leq s$ implies $-s \leq -w$, the number $-w$ is an upper bound of S' . By completeness $u = \sup S'$ exists. We claim $-u = \inf S$.

$-u$ is a lower bound: $-s \leq u$ for all $s \in S$, so $-u \leq s$.

$-u$ is the greatest one: if w is any lower bound of S , then $-w$ is an upper bound of S' , so $u \leq -w$, i.e. $w \leq -u$.

■

Example 1.5.2

The set $S = \left\{ \left(1 + \frac{1}{n}\right)^n : n \in \mathbb{N} \right\}$ satisfies $1 < x < 3$ for all $x \in S$; by completeness, it has a supremum.

Proof

By the binomial theorem,

$$\left(1 + \frac{1}{n}\right)^n = 1 + \binom{n}{1}\frac{1}{n} + \sum_{k=2}^n \binom{n}{k}\frac{1}{n^k} > 1.$$

For $0 \leq k \leq n$, $\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k!} \leq \frac{n^k}{k!}$, and $k! \geq 2^{k-1}$ for $k \geq 2$ (each factor $2, \dots, k$ is at least 2). Therefore

$$\left(1 + \frac{1}{n}\right)^n \leq \sum_{k=0}^n \frac{1}{k!} < 1 + 1 + \sum_{k=2}^{\infty} \frac{1}{2^{k-1}} = 3.$$

■

b. The Archimedean property

Theorem 1.5.3 (Archimedean property)

For every $x \in \mathbb{R}$ there is $n_x \in \mathbb{N}$ with $x \leq n_x$. In other words, $\mathbb{N}$ is not bounded above in $\mathbb{R}$.

The proof uses two ingredients: completeness, to form $u = \sup \mathbb{N}$, and the closure of $\mathbb{N}$ under $n \mapsto n + 1$, to step beyond u .

Proof

Suppose some $x \in \mathbb{R}$ satisfies $x > n$ for all $n \in \mathbb{N}$. Then $\mathbb{N}$ is nonempty and bounded above, so by completeness $u = \sup \mathbb{N}$ exists in $\mathbb{R}$. Since $u - 1 < u$ and u is the *least* upper bound, $u - 1$ is not an upper bound of $\mathbb{N}$ (Lemma 1.4.3 with $\epsilon = 1$): there is $m \in \mathbb{N}$ with $u - 1 < m$. Then $u < m + 1$. But $m + 1 \in \mathbb{N}$ and u is an upper bound of $\mathbb{N}$, so $m + 1 \leq u$ — a contradiction. ■

Corollary 1.5.4

- (a) If $S = \{1/n : n \in \mathbb{N}\}$, then $\inf S = 0$.
- (b) If $t > 0$, there is $n_t \in \mathbb{N}$ with $0 < 1/n_t < t$.
- (c) If $y > 0$, there is $n_y \in \mathbb{N}$ with $n_y - 1 \leq y < n_y$.

Proof

We prove (b) first, since (a) uses it.

(b) Choose $m \in \mathbb{N}$ with $1/t \leq m$ and put $n_t = m + 1$. Then $n_t > 1/t$, i.e. $0 < 1/n_t < t$.

(a) 0 is a lower bound of S . If $b > 0$, part (b) gives n with $1/n < b$, so b is not a lower bound. Hence $\inf S = 0$.

(c) We want the *first* natural number to the right of y , so we consider all candidates,

$$A = \{n \in \mathbb{N} : y < n\}.$$

By the Archimedean property $A \neq \emptyset$, so by well-ordering (Theorem 1.2.3) A has a least element n_y , and $y < n_y$ because $n_y \in A$. If we had $n_y - 1 > y$, then $n_y - 1 > 0$, so $n_y - 1 \in \mathbb{N}$ and hence $n_y - 1 \in A$, contradicting the minimality of n_y . Thus $n_y - 1 \leq y < n_y$. ■

Remark

Part (c) is a *minimum* argument, not a supremum argument. The set A is unbounded above, so $\sup A$ does not exist; what we need is the least element of A , and well-ordering supplies it for any nonempty subset of $\mathbb{N}$, bounded or not.

c. Existence of $\sqrt{2}$

Proposition 1.5.5

There is a real number x with $x^2 = 2$.

We define x as the supremum of the set S of all nonnegative t with $t^2 < 2$, i.e. of all “approximations of $\sqrt{2}$ from below”. Then we rule out both alternatives to $x^2 = 2$:

- if $x^2 < 2$, then x is too small: $x + \frac{1}{n}$ is still in S , so x is not an upper bound;
- if $x^2 > 2$, then x is too large: $x - \frac{1}{n}$ is still an upper bound, so x is not the least one.

In both cases n is chosen by the Archimedean property.

Proof

Let $S = \{t \in \mathbb{R} : t \geq 0, t^2 < 2\}$. Then $1 \in S$, and 2 is an upper bound of S (if $t > 2$ then $t^2 > 4$). By completeness $x = \sup S$ exists, and $x \geq 1$.

Case $x^2 < 2$. For every $n \in \mathbb{N}$, since $1/n^2 \leq 1/n$,

$$\left(x + \frac{1}{n}\right)^2 = x^2 + \frac{2x}{n} + \frac{1}{n^2} \leq x^2 + \frac{2x+1}{n}.$$

As $2 - x^2 > 0$, the Archimedean property gives n with $\frac{2x+1}{n} < 2 - x^2$. Then $\left(x + \frac{1}{n}\right)^2 < 2$, so $x + \frac{1}{n} \in S$ although $x + \frac{1}{n} > x$. This contradicts that x is an upper bound of S .

Case $x^2 > 2$. For every $n \in \mathbb{N}$,

$$\left(x - \frac{1}{n}\right)^2 = x^2 - \frac{2x}{n} + \frac{1}{n^2} > x^2 - \frac{2x}{n}.$$

As $x^2 - 2 > 0$, the Archimedean property gives n with $\frac{2x}{n} < x^2 - 2$. Then $\left(x - \frac{1}{n}\right)^2 > 2 > t^2$ for every $t \in S$. Since $x - \frac{1}{n} \geq 0$ and $s \mapsto s^2$ is strictly increasing on $[0, \infty)$, this gives $t < x - \frac{1}{n}$ for all $t \in S$. So $x - \frac{1}{n}$ is an upper bound of S smaller than x , contradicting $x = \sup S$.

Hence $x^2 = 2$. ■

d. Density of $\mathbb{Q}$ in $\mathbb{R}$ **Theorem 1.5.6**

If $x, y \in \mathbb{R}$ and $x < y$, there is $r \in \mathbb{Q}$ with $x < r < y$.

Let $0 < x < y$ first. We look for $r = m/n$. Choose n so large that the grid of multiples of $1/n$ is finer than the gap $y - x$; then some grid point must fall in (x, y) , and m/n is the first grid point to the right of x .

Proof

Case $0 < x < y$. By the Archimedean property choose $n \in \mathbb{N}$ with $1/n < y - x$, i.e. $nx + 1 < ny$. By Corollary 1.5.4(c) applied to $nx > 0$, there is $m \in \mathbb{N}$ with

$m - 1 \leq nx < m$. Then

$$nx < m \leq nx + 1 < ny,$$

and dividing by n gives $x < m/n < y$.

Case $x \leq 0$. Choose $k \in \mathbb{N}$ with $-x < k$. Then $0 < x + k < y + k$, so by the first case there is $q \in \mathbb{Q}$ with $x + k < q < y + k$, and $r = q - k \in \mathbb{Q}$ satisfies $x < r < y$. ■

Corollary 1.5.7

If $x < y$, there is an irrational z with $x < z < y$.

Proof

Apply Theorem 1.5.6 to $x/\sqrt{2} < y/\sqrt{2}$ to obtain a rational $r \neq 0$ between them (if $x < 0 < y$, take r between 0 and $y/\sqrt{2}$). Then $z = \sqrt{2}r$ satisfies $x < z < y$, and $z \notin \mathbb{Q}$, since otherwise $\sqrt{2} = z/r$ would be rational. ■

Corollary 1.5.8

If $x < y$, there are infinitely many rationals between x and y .

Proof

Choose $r_1 \in \mathbb{Q} \cap (x, y)$; having chosen r_k , choose $r_{k+1} \in \mathbb{Q}$ with $r_k < r_{k+1} < y$. The numbers $r_1 < r_2 < \dots$ are distinct rationals in (x, y) . ■

We can now make precise the claim that $\mathbb{Q}$ is not complete. The set T below is the rational analogue of the set S used to construct $\sqrt{2}$; its supremum “ought to be” $\sqrt{2}$, which is not in $\mathbb{Q}$.

Proposition 1.5.9

The set $T = \{r \in \mathbb{Q} : r^2 < 2\}$ is nonempty and bounded above in $\mathbb{Q}$, but has no least upper bound in $\mathbb{Q}$.

Proof

$1 \in T$, and 2 is an upper bound. Let $r \in \mathbb{Q}$ be any upper bound of T ; we find a smaller rational upper bound. Note $r \geq 1$ and $r \neq \sqrt{2}$ (Theorem 1.1.1).

If $r < \sqrt{2}$, Theorem 1.5.6 gives $q \in \mathbb{Q}$ with $r < q < \sqrt{2}$; then $q > 0$ and $q^2 < 2$, so $q \in T$ and $q > r$, contradicting that r is an upper bound. Hence $r > \sqrt{2}$, and Theorem 1.5.6 gives $q \in \mathbb{Q}$ with $\sqrt{2} < q < r$. For $t \in T$ we have $t^2 < 2 < q^2$ with $q > 0$, so $t < q$: thus q is a rational upper bound of T smaller than r . ■

e. Countable sets

Are there more rational or irrational numbers? To give the question a meaning, we compare sets by means of bijections.

Definition 1.5.10

Sets A and B have the same *cardinality*, $|A| = |B|$, if there is a bijection (a map that is injective and surjective) $f : A \rightarrow B$. A set S is

- (1) *finite* if $S = \emptyset$ or there is a bijection $\{1, \dots, n\} \rightarrow S$ for some $n \in \mathbb{N}$;
- (2) *countably infinite* if there is a bijection $\mathbb{N} \rightarrow S$;
- (3) *countable* if it is finite or countably infinite, and *uncountable* otherwise.

Example 1.5.11

(a) $f(n) = 2n$ is a bijection from $\mathbb{N}$ onto the even natural numbers.

(b) $g(1) = 0, g(2k) = k, g(2k + 1) = -k$ ($k \in \mathbb{N}$) is a bijection $\mathbb{N} \rightarrow \mathbb{Z}$. Thus both sets are countably infinite: a set can have the same cardinality as a proper subset of itself.

2 Sequences and Limits

The completeness of $\mathbb{R}$ is useful only once we can speak of numbers being approached. The simplest setting for this is a sequence, and the central notion is that of its limit.

2.1 The Limit of a Sequence

a. Definition and first examples

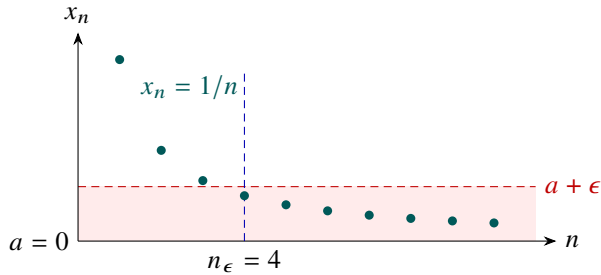
A *sequence* of real numbers is a function $X : \mathbb{N} \rightarrow \mathbb{R}$. We write $x_n = X(n)$ for its n th term and denote the sequence by $\{x_n\}_{n=1}^{\infty}$ or simply $\{x_n\}$.

Definition 2.1.1

A sequence $\{x_n\}$ *converges* to $a \in \mathbb{R}$, written $\lim_{n \rightarrow \infty} x_n = a$ or $x_n \rightarrow a$, if

$$(\forall \epsilon > 0)(\exists n_\epsilon \in \mathbb{N})(\forall n \geq n_\epsilon) [|x_n - a| < \epsilon].$$

In words: for every tolerance $\epsilon > 0$, all sufficiently late terms lie within ϵ of a . The index n_ϵ is allowed to depend on ϵ (smaller tolerances usually require later indices), and finitely many early terms may lie anywhere. In the figure, $x_n = 1/n$, $a = 0$, $\epsilon = 0.3$: every term from $n_\epsilon = 4$ on lies in the shaded band.



The interval

$$V_\epsilon(a) = \{x \in \mathbb{R} : |x - a| < \epsilon\} = (a - \epsilon, a + \epsilon)$$

is called the ϵ -neighbourhood of a . Thus $x_n \rightarrow a$ if and only if every ϵ -neighbourhood of a contains all terms of the sequence from some index on.

To prove $x_n \rightarrow a$ from the definition, one fixes $\epsilon > 0$, estimates $|x_n - a|$ from above by something simple, and chooses n_ϵ — usually by the Archimedean property — to make that bound less than ϵ .

Example 2.1.2

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Proof

Given $\epsilon > 0$, choose n_ϵ with $1/n_\epsilon < \epsilon$ (Corollary 1.5.4(b)). For $n \geq n_\epsilon$, $|\frac{1}{n} - 0| =$

$$\frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon.$$

Example 2.1.3

$$\lim_{n \rightarrow \infty} \frac{4n+3}{n+1} = 4.$$

Proof

Given $\epsilon > 0$, choose n_ϵ with $1/n_\epsilon < \epsilon$. For $n \geq n_\epsilon$,

$$\left| \frac{4n+3}{n+1} - 4 \right| = \frac{1}{n+1} < \frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon.$$

b. Uniqueness of the limit**Proposition 2.1.4**

If $x, y \in \mathbb{R}$ and $|x - y| < \epsilon$ for every $\epsilon > 0$, then $x = y$.

Proof

If $x \neq y$, take $\epsilon = |x - y|/2 > 0$; then $|x - y| > \epsilon$.

Theorem 2.1.5

A sequence has at most one limit.

If $x_n \rightarrow a$ and $x_n \rightarrow b$, a late enough term is close to both a and b , so by the triangle inequality a and b are close to each other.

Proof

Let $\epsilon > 0$. Choose n_1 with $|x_n - a| < \epsilon/2$ for $n \geq n_1$ and n_2 with $|x_n - b| < \epsilon/2$ for $n \geq n_2$. For $n \geq \max\{n_1, n_2\}$,

$$|a - b| \leq |a - x_n| + |x_n - b| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Since $\epsilon > 0$ was arbitrary, $a = b$ by Proposition 2.1.4.

c. Divergence and boundedness

A sequence that does not converge to any real number is said to *diverge*.

Example 2.1.6

The sequence $0, 1, 0, 1, \dots$ ($x_n = 0$ for odd n , $x_n = 1$ for even n) diverges: both values recur arbitrarily late, so the terms cannot all stay near one number.

Proof

Suppose $x_n \rightarrow a$ and take $\epsilon = 1/2$. Choosing an odd and an even index $n \geq n_\epsilon$ gives $|0 - a| < \frac{1}{2}$ and $|1 - a| < \frac{1}{2}$, i.e. $-\frac{1}{2} < a < \frac{1}{2}$ and $\frac{1}{2} < a < \frac{3}{2}$, which is impossible. ■

Definition 2.1.7

A sequence $\{x_n\}$ is *bounded* if there is $M \in \mathbb{R}$ with $|x_n| \leq M$ for all $n \in \mathbb{N}$.

Theorem 2.1.8

Every convergent sequence is bounded.

Convergence controls all terms from some index on; only finitely many terms remain, and a finite set is bounded.

Proof

Let $x_n \rightarrow a$. With $\epsilon = 1$, choose n_1 such that $|x_n - a| < 1$ for $n \geq n_1$; then $|x_n| \leq |x_n - a| + |a| < 1 + |a|$ for these n . Hence

$$M = \max \{|x_1|, \dots, |x_{n_1-1}|, 1 + |a|\}$$

bounds every term (if $n_1 = 1$, take $M = 1 + |a|$). ■

The converse is false: the sequence of Example 2.1.6 is bounded but divergent. On the other hand, by Theorem 2.1.8 an *unbounded* sequence must diverge.

Example 2.1.9

The following sequences are unbounded, hence divergent.

- (a) $a_n = \frac{n^2+1}{n}$. Here $|a_n| = n + \frac{1}{n}$. Given $M > 0$, choose $n_M \in \mathbb{N}$ with $n_M > M$; then $|a_{n_M}| > M$.
 (b) $a_n = (-1)^n n$. Here $|a_n| = n$, and again $|a_{n_M}| > M$ whenever $n_M > M$.

d. Infinite limits**Definition 2.1.10**

A sequence $\{x_n\}$ *diverges to $+\infty$* , written $\lim_{n \rightarrow \infty} x_n = +\infty$, if for every $M > 0$ there is $n_M \in \mathbb{N}$ such that $x_n > M$ for all $n \geq n_M$. It diverges to $-\infty$ if for every $M > 0$ there is n_M such that $x_n < -M$ for all $n \geq n_M$.

Such a sequence is unbounded and therefore *divergent*; the notation $\lim x_n = \pm\infty$ only records the manner in which it diverges.

Example 2.1.11

$$\lim_{n \rightarrow \infty} \frac{n^2}{1-n} = -\infty.$$

Proof

For $n \geq 2$ we have $0 < n-1 < n$, so $\frac{n^2}{n-1} > \frac{n^2}{n} = n$, i.e. $\frac{n^2}{1-n} < -n$. Given $M > 0$, choose $n_M \in \mathbb{N}$ with $n_M \geq 2$ and $n_M > M$. For $n \geq n_M$,

$$\frac{n^2}{1-n} < -n \leq -n_M < -M.$$

■

Example 2.1.12

$$\lim_{n \rightarrow \infty} (\sqrt{n} + 1) = +\infty.$$

Proof

Given $M > 0$, we must make $\sqrt{n} > M - 1$ for all late n . Choose $n_M \in \mathbb{N}$ with $n_M > M^2$. For $n \geq n_M$, $\sqrt{n} \geq \sqrt{n_M} > M > M - 1$, so $\sqrt{n} + 1 > M$.

■

e. Limits and arithmetic operations**Theorem 2.1.13**

Let $\lim_{n \rightarrow \infty} x_n = a$ and $c \in \mathbb{R}$. Then

- (1) $\lim_{n \rightarrow \infty} c = c$ (the constant sequence);
- (2) $\lim_{n \rightarrow \infty} cx_n = ca$.

Proof

(1) $|c - c| = 0 < \epsilon$ for every n .

(2) If $c = 0$, this is (1). If $c \neq 0$, given $\epsilon > 0$ choose n_ϵ with $|x_n - a| < \epsilon/|c|$ for $n \geq n_\epsilon$; then $|cx_n - ca| = |c| |x_n - a| < \epsilon$.

■

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